

Joshua Sakthivel Raju

Developer ~ Engineer

+91 99304 40108

Mumbai, India

joshua.raju2604@gmail.com

github.com/JoshuaR26

linkedin/joshua-sakthivel-raju

joshuar26.github.io/Portfolio/

SUMMARY

Passionate AI/ML engineer with a strong background in deep learning, computer vision, and natural language processing. Skilled in Python, TensorFlow, PyTorch, and various ML libraries. Demonstrates excellent problem-solving, research, collaboration, and leadership abilities, with experience leading teams and projects to success. Seeking a challenging role to develop cutting-edge AI solutions.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, Java, SQL, React.JS, HTML, SCSS

Technologies: Deep Learning, Computer Vision, Machine Learning, Natural Language Processing, Web Development, Game Development

PROJECTS

Surgical Tool Segmentation

Python, TensorFlow, OpenCV

Deep Learning, Computer Vision

- Engineered advanced U-Net models that achieved a remarkable Intersection over Union (IoU) score of 94.8%, leading to substantial advancements in both surgical precision and team workflow effectiveness.

InTellik

React, JavaScript, Python

Natural Language Processing, Web Application

- Winner of Dataset 2024 National-Level Hackathon with more than 300+ participants.
- Fine-tuned Llama 3.1 (8B parameters) and implemented Retrieval-Augmented Generation (RAG) to leverage customer interaction histories for personalized service recommendations.

sMile

HTML, SCSS, JavaScript, PHP

Web Application

- Winner of ACM (Association for Computing Machinery) WebDex 2024 Hackathon with more than 250+ participants.
- Built a platform addressing 4 Sustainable Development Goals, connecting volunteers with organizations for impactful collaboration.

EXPERIENCE

Machine Learning Researcher

Feb 2024 – Present

University of Pretoria, South Africa

- Led the Augmentation Team in collaboration with the University of Pretoria's Data Science for Social Good Lab under Prof. Vukosi Marivate.
- Worked on Text augmentation for multilingual NLP models (African languages) with 150,000+ users in the base repository.

MLOps & Data Pipeline Developer

Jun 2024 – Sep 2024

Logeuler Pvt. Ltd.

- Constructed and managed robust data pipelines and workflows to streamline model training, evaluation, and deployment, and monitor model performance within an ERP (Enterprise Resource Planning) environment.

Generative AI Vision Intern

Jul 2024 – Sep 2024

V3 Analytics Pvt. Ltd.

- Contributed to project development by implementing advanced techniques in artificial intelligence and enhancing proficiency in image generation algorithms.

Google Data Science Dept. Lead

Aug 2024 – Present

Google Developer Groups on Campus

Promoted

- Championed innovative solutions by implementing predictive analytics models, and directly impacting the successful launch of 3 major club initiatives, including MoUs with industrial companies.

Office Bearer

Sep 2024 – Present

VITeach Non-Profit Organization

Promoted

- Effectively managed a diverse team of 250+ volunteers while fostering partnerships with local schools and NGOs to expand outreach and promote education for underprivileged communities.

ACHIEVEMENTS

Dataset 2024 National-Level Hackathon Winner (1st Place) <i>Nokia, Prodapt, Engagements and Collaborations committee of Data Science</i>	Technical Member 19 Dec 2024 (36hr)
ACM WebDex 2024 Hackathon Winner (1st Place) <i>Association for Computing Machinery</i>	Team Lead 22 Mar 2024 (24hr)
Game-A-Thon Hackathon Winner (1st Place) <i>School of Computer Science and Engineering, VIT, Chennai</i>	Lead Artist 27 Sep 2023

PUBLICATIONS

• Joshua S Raju, Krishna, K.V Hitesh, Adityasinh Rana, and Thomas Abraham J.V "VocalVisageNet: A Spatio-Temporal Transformer approach for Lip-Reading" (2025) (in press)	
• Krishna, K.V Hitesh, Joshua S Raju, Dr. Suganeshwari G, and Dr. Sudha A "A Shallow ResNet-Based GAN for Super-Resolution Tasks" (2025) (in press)	
• Krishna, K.V Hitesh, Joshua S Raju, and Kanimozhi S. "TamilHRNet: Three-stage segmentation network for Tamil handwritten document digitalization." (2024) (in press)	
• Riddhiman Raguraman, Joshua Sakthivel Raju, and Subbulakshimi P. "Adaptive NPC in Serious Games Using Artificial Intelligence" (2024) (in press)	

EDUCATION

B.Tech in Computer Science with Specialization in Artificial Intelligence & Machine Learning <i>Vellore Institute of Technology, Chennai, Tamil Nadu</i>	Sep 2022 – Present
Sr. Secondary, CBSE(XII) <i>Delhi Public School, Navi Mumbai, Maharashtra</i>	2020 – 2022
Secondary, CBSE(X) <i>Delhi Public School, Navi Mumbai, Maharashtra</i>	2010 – 2020

SPOKEN LANGUAGES

Professional Proficiency : English, Hindi	
Elementary Proficiency : Telugu, Tamil, Spanish	